

Gerrit Merge Conflict Resolver

User's Manual

CS 5150 — Group 17 — Spring 2026

1. Overview

The Gerrit Merge Conflict Resolver is a browser-based, three-panel interface embedded directly in Gerrit that allows developers to resolve merge conflicts without leaving their code review workflow. When a rebase produces conflicts, Gerrit can now open a dedicated merge editor that shows the Current branch, the Incoming branch, and an editable Result pane side by side, with conflicting regions highlighted in distinct colors. Once all conflicts are resolved, the result is saved as a new patch set using Gerrit's existing inline-edit flow.

This manual covers the full workflow for the primary user role: a developer (change owner) who needs to resolve a merge conflict on one of their Gerrit changes.

2. Prerequisites

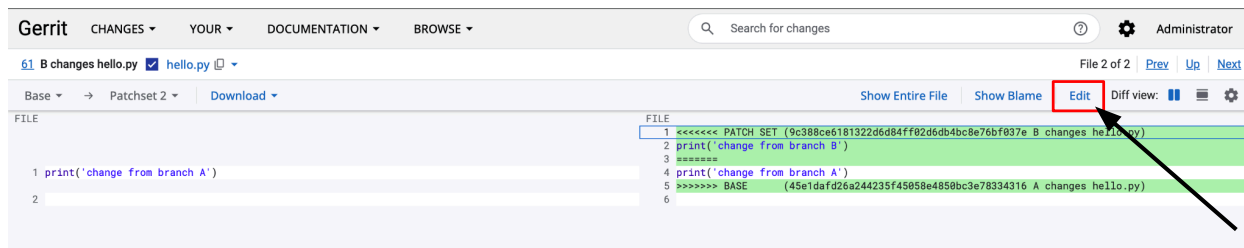
- You have a Gerrit account and are the owner of a change that has a merge conflict (i.e., a rebase was attempted and failed with conflicts).
- You are using a modern browser (Chrome or Firefox recommended). The merge editor uses Polymer web components and requires JavaScript to be enabled.
- You have permission to edit the change (change owner or equivalent role).

3. Navigating to the Merge Conflict Editor

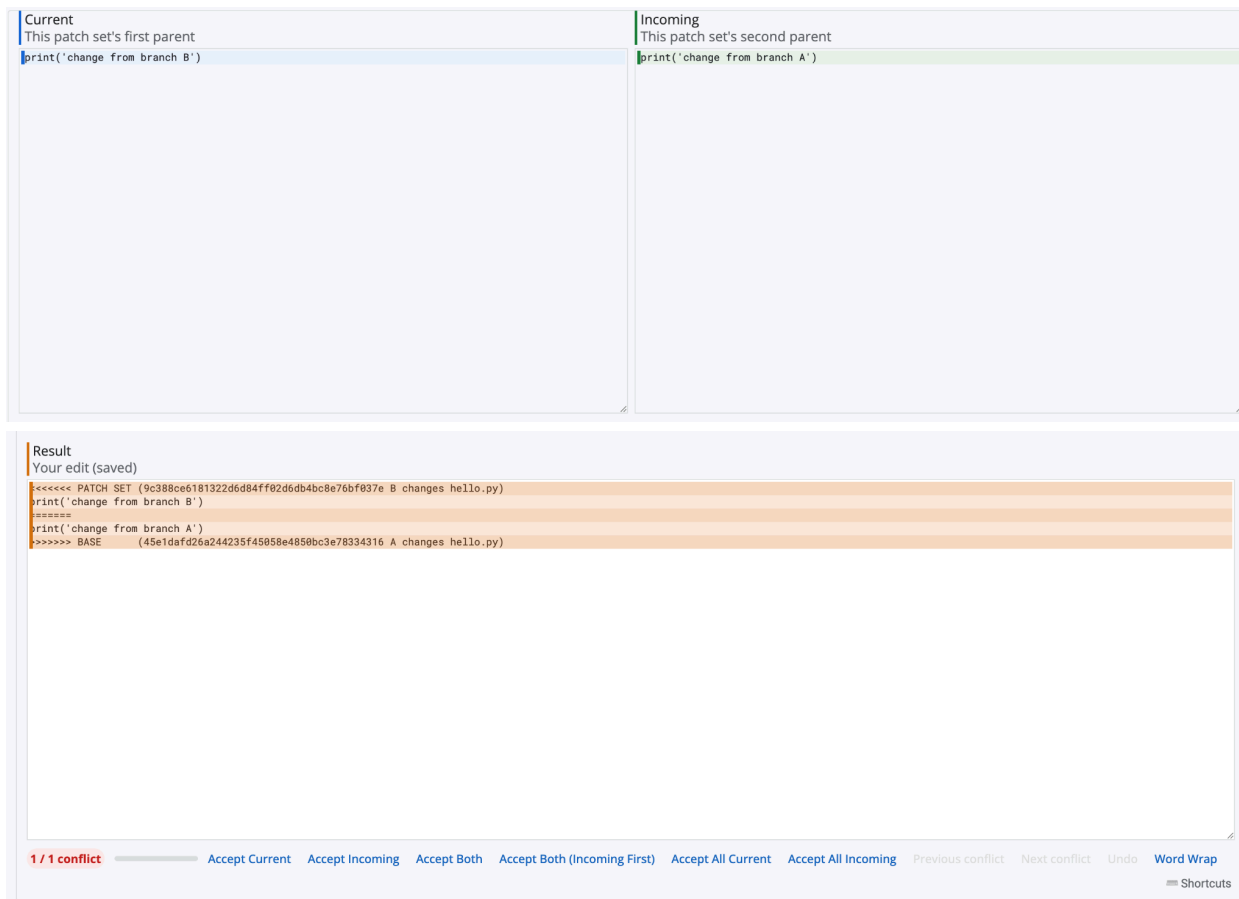
When a change has an unresolved merge conflict, Gerrit will indicate this on the change page. To open the merge editor:

1. Open the change in Gerrit by navigating to its change page.
2. Look for the “Merge Conflict” indicator or banner near the top of the change page.
3. Choose a file with conflicts in it.
4. Click the “Edit” button in the top right. This opens the gr-merge-editor view for the conflicted file.
5. If the change has multiple conflicted files, each file must be resolved individually. Select the file you wish to resolve from the file list.

Note: The merge editor is only available for changes that contain git conflict markers (<<<<<< / ===== / >>>>>>). Changes without conflicts will not show this option.



4. Understanding the Three-Panel Interface



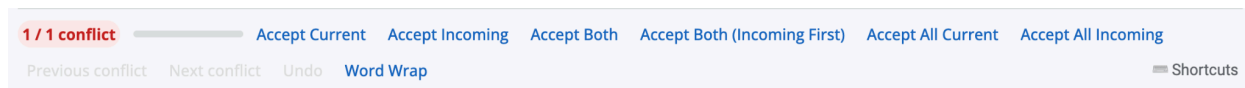
The merge editor displays three vertical panels:

Panel	Description
Current (left)	Shows the content from your branch; the version you were working on before the rebase. Read-only.
Incoming (right)	Shows the content from the branch being rebased onto. Read-only.
Result (center)	An editable pane where you compose the final resolved version of the file. This is what will be saved as the new patch set.

Conflicting regions are highlighted with color-coded backgrounds: blue for the Current side and green for the Incoming side. Non-conflicting lines are shown in all panels without highlighting.

A progress indicator at the bottom of the editor shows how many conflicts remain unresolved (e.g., “2/5 conflict”).

5. Resolving Conflicts



5.1 Navigating Between Conflicts

Use the Previous and Next conflict buttons in the bottom toolbar to jump between conflicting regions. You can also use keyboard shortcuts (see Section 7).

5.2 Accepting a Resolution

For each conflict, you have four resolution options available via the buttons at the bottom of the editor:

Action	Effect
Accept Current	Copies the Current (left) version of the conflict into the Result pane, discarding the Incoming version.
Accept Incoming	Copies the Incoming (center) version into the Result pane, discarding the Current version.
Accept Both	Appends both versions sequentially into the Result pane (Current first, then Incoming). There is a separate button that appends both versions with Incoming first, then Current.
Accept All (Current / Incoming)	Applies Accept Current or Accept Incoming to every remaining unresolved conflict at once. Use with care.

After accepting a resolution, the conflict marker arrows are removed from the Result pane and the progress counter updates.

5.3 Manual Editing

You can also type directly in the Result pane at any time. This is useful when neither the Current nor Incoming version is correct on its own and you need to write a merged version by hand. Click anywhere in the Result pane and edit the text as you would in any text editor.

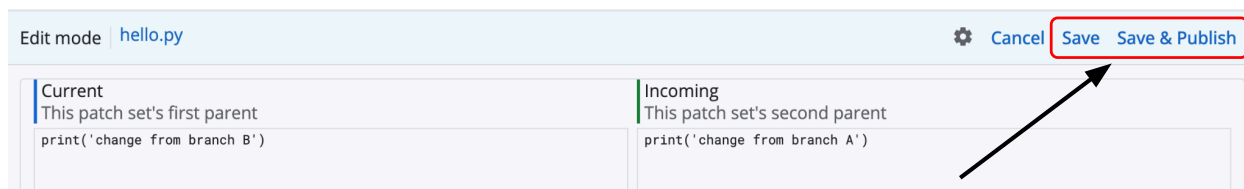
Note: *Manual edits are not automatically re-parsed as resolved conflicts. After editing a conflicting region manually, use the Accept Current or Accept Incoming button (or remove the conflict markers yourself) to mark that conflict as resolved.*

5.4 Undoing a Resolution

To undo the most recent resolution action, click the Undo button in the toolbar. Undo steps through your resolution history one action at a time. You can undo multiple times to step back through previous decisions.

Note: *Undo applies to accept/resolution actions. Direct edits in the Result pane can be undone using the browser's native undo (Ctrl+Z / Cmd+Z) within the text area.*

6. Saving and Publishing the Resolution

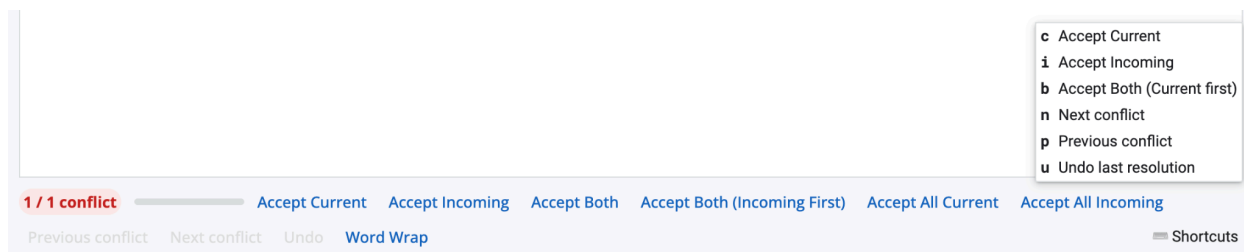


Once all conflicts are resolved (the progress indicator reads “All conflicts resolved”), you can save the result:

6. Click the Save button at the top of the editor. This commits the resolved file content as a new patch set on the change using Gerrit’s inline-edit flow.
7. Gerrit will create a new patch set containing your resolved file. You will be redirected to the change page where the new patch set is visible.
8. If the change has additional conflicted files, repeat the process for each file before publishing.
9. Once all files are resolved and saved, publish the new patch set for review as you would any other patch set.

Note: Saving does not immediately publish the patch set for review. It stages the resolved content as an unpublished edit. You must explicitly publish the edit from the change page to make it visible to reviewers or use the Save & Publish button.

7. Keyboard Shortcuts



All major actions in the merge editor are accessible via keyboard shortcuts. To view the full shortcut reference, press the button on the bottom right called Shortcuts.

Shortcut	Action
c	Accept Current for the active conflict
i	Accept Incoming for the active conflict
b	Accept Both for the active conflict
n	Jump to Next conflict
p	Jump to Previous conflict
u	Undo last resolution

Note: Gerrit’s own keyboard shortcuts are accessible by pressing “?”. It gives access to global shortcuts and does not conflict with the merge editor keyboard shortcuts.

8. Display Options

8.1 Dark Mode

If the user is in Dark Mode (toggle in setting to switch between light and dark themes), the UI is updated to white text on all three panels and the toolbar for visibility.

8.2 Word Wrap

Click the Word Wrap toggle in the bottom toolbar to enable or disable line wrapping in all panels. When disabled, long lines require horizontal scrolling. When enabled, lines wrap at the panel boundary.

9. Troubleshooting

Issue	Resolution
The “Resolve Conflicts” option does not appear on the change page.	Verify the change contains git conflict markers (<<<<<<< lines). If the rebase was allowed with conflicts but markers were not committed, the editor will not activate. Check the raw file content.
The Result pane shows conflict markers after accepting a resolution.	This can occur if Accept Current/Incoming was applied but the markers were not fully removed. Manually delete any remaining <<<<<<< / ===== / >>>>>>> lines from the Result pane.
Progress counter does not update after a manual edit.	Manual edits are not automatically detected as conflict resolutions. Use an Accept button or manually remove the conflict marker lines to update the counter.
Save button is disabled.	Ensure all conflict markers have been resolved (progress indicator reads “All conflicts resolved”). If markers remain, the save path will not proceed.
Reference pane (base column) shows an error or is empty.	The base (common ancestor) column is partially implemented. If it fails to load, the Current and Incoming panels are unaffected and resolution can proceed normally.
Keyboard shortcuts do not work.	Ensure the “Result” panel is not selected and try again.

10. Glossary

Term	Definition
Merge conflict	A situation where two branches have made incompatible changes to the same lines of a file, preventing automatic merging.
Current	The version of the file on your branch (the branch being rebased).
Incoming	The version of the file on the branch being rebased onto.
Result pane	The editable output pane in the merge editor where the resolved content is composed.
Patch set	A revision of a Gerrit change. Saving a resolved conflict creates a new patch set.
Inline edit	Gerrit's mechanism for making small edits directly in the browser without a local checkout. The merge editor uses this flow to save resolutions.
Conflict marker	The <<<<<<< / ===== / >>>>>>> lines inserted by Git to delimit conflicting regions in a file.